

EXPERIMENT-5:

Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python.

Code:

```
# =====  
# K-Nearest Neighbours (KNN)  
# =====  
  
import pandas as pd  
from google.colab import files  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import LabelEncoder  
from sklearn.neighbors import KNeighborsClassifier  
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report  
  
# Upload CSV file  
uploaded = files.upload()  
  
# Read CSV  
filename = next(iter(uploaded))  
df = pd.read_csv(filename)  
  
print("First 5 Rows:")  
print(df.head())  
  
# Remove Id column if present  
if "Id" in df.columns:  
    df = df.drop("Id", axis=1)  
  
# Features  
X = df.iloc[:, :-1]
```

```
# Target
y = df.iloc[:, -1]

# Encode target labels
le = LabelEncoder()
y = le.fit_transform(y)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create KNN model
knn = KNeighborsClassifier(n_neighbors=5)

# Train model
knn.fit(X_train, y_train)

# Predict
y_pred = knn.predict(X_test)

# Accuracy
print("\nAccuracy:", accuracy_score(y_test, y_pred) * 100)

# Confusion Matrix
print("\nConfusion Matrix")
print(confusion_matrix(y_test, y_pred))
```

```

# Classification Report

print("\nClassification Report")

print(classification_report(y_test, y_pred, target_names=le.classes_))

# Predict a new sample

sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = knn.predict(sample)

print("\nPredicted Species:", le.inverse_transform(prediction)[0])

```

Output:

Choose Files

Iris[1].csv

... **Iris[1].csv**(text/csv) - 5107 bytes, last modified: 8/3/2026 - 100% done

Saving Iris[1].csv to Iris[1] (1).csv

First 5 Rows:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Accuracy: 100.0

Confusion Matrix

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Classification Report

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Predicted Species: Iris-setosa